EXP.NO: 6	Write a Servlet to demonstrate session tracking using HttpSession.

AIM:

To demonstrate session tracking using HttpSession in a Servlet-based web application. The exercise involves implementing a simple login system where a user can log in with their username and password, and their session will be tracked throughout their interaction with the web application.

ALGORITHM:

1. Input:

- The user will enter their **username** and **password** in a **login form**.
- The servlet will validate the credentials and create an **HTTP session** for the user if authentication is successful.

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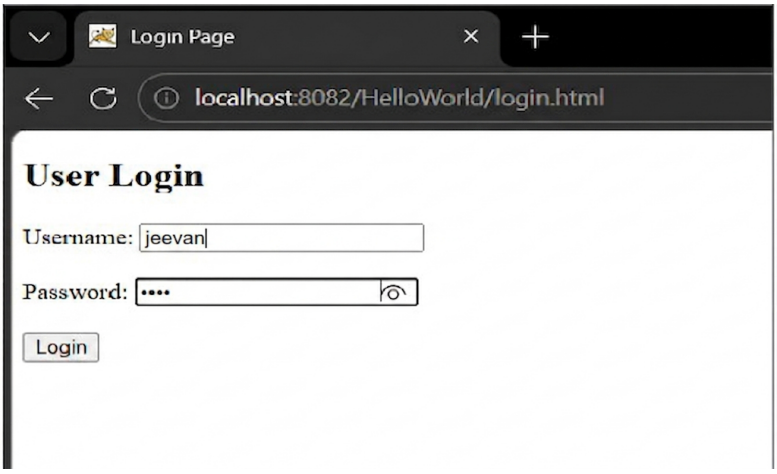
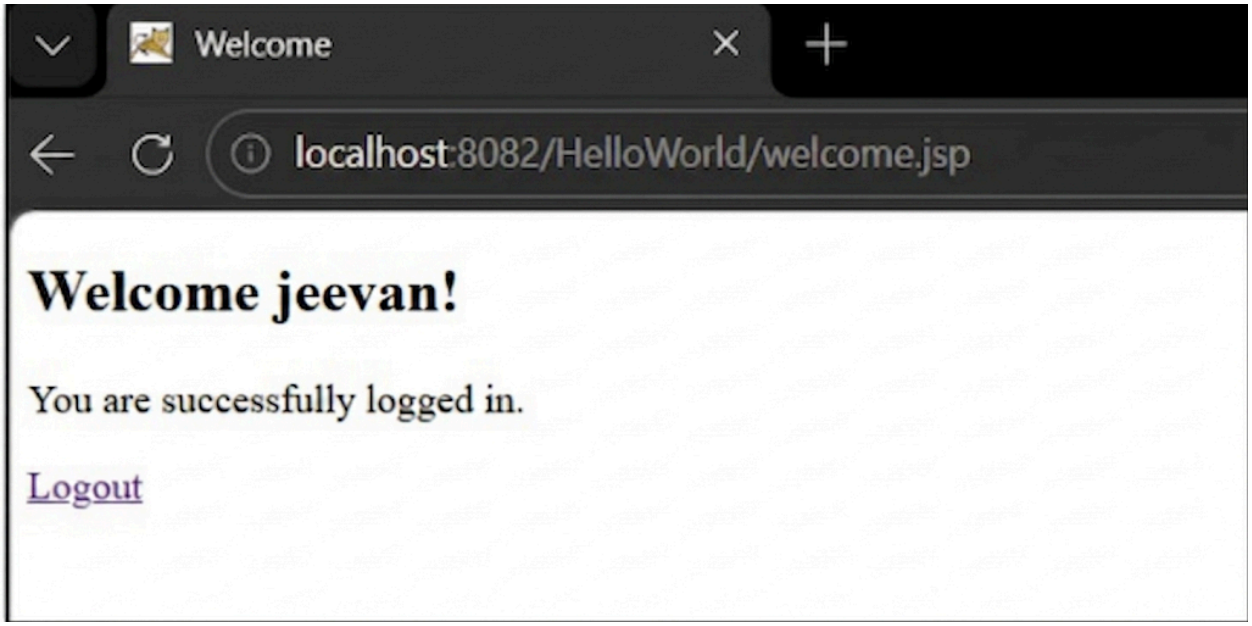
2. Steps for Implementing Session Tracking with HttpSession:

1. **Create the Login Form (HTML):**
   - Create an HTML form with two input fields for **username** and **password**.
   - Provide a **submit button** to send the data to the servlet for authentication.
2. **Create the Login Servlet:**
   - Write a Java servlet class to handle **login** requests.
   - The servlet will:
     - Retrieve the **username** and **password** from the form.
     - Authenticate the user by comparing the credentials to a stored value (e.g., hardcoded or from a database).
     - Create a new **HTTP session** for the user if the credentials are valid.
     - Set session attributes such as the user's **username** to track the session.
3. **Validate User Credentials:**
   - In the servlet, check if the **username** and **password** match the expected values.
   - If valid, create a session and store the **username** in the session.
   - If invalid, redirect the user back to the login page with an error message.

4. **Create and Track User Session:**
    - Upon successful login, use the HttpSession object to store the session data.
    - Use the session.setAttribute("username", username) method to save the **username**.
    - Optionally, set session timeout using session.setMaxInactiveInterval(time_in_seconds) to manage session expiration.
5. **Redirect to a Protected Page:**
    - After a successful login, the user will be redirected to a **protected page** (e.g., a dashboard or home page).
    - The protected page should check if the user is authenticated by checking the **session attribute**.
6. **Logout and Session Invalidation:**
    - Provide a **logout feature** that invalidates the session using session.invalidate() when the user logs out.
    - After logout, redirect the user back to the login page.
7. **Session Expiration:**
    - If the session expires or the user manually logs out, the session will be invalidated, and the user will be redirected to the login page.

Output:





HTML File (login.html):

```
<!DOCTYPE html>

<html>

<head>

<title>Login Page</title>

</head>

<body>

<h2>User Login</h2>

<form action="login" method="post">

Username: <input type="text" name="username" required><br><br>

Password: <input type="password" name="password" required><br><br>

<input type="submit" value="Login">

</form>

</body>
```

```
</html>
```

Java File (LoginServlet.java) :

```
import java.io.IOException;
import java.io.PrintWriter;

import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;

@WebServlet("/login")
public class LoginServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        String username = request.getParameter("username");
        String password = request.getParameter("password");
```

```
        if ("admin".equals(username) && "1234".equals(password)) {

            HttpSession session = request.getSession();
            session.setAttribute("username", username);

            session.setMaxInactiveInterval(300);

            response.sendRedirect("welcome.jsp");

        } else {
            out.println("<h3 style='color:red'>Invalid Username or Password</h3>");
            request.getRequestDispatcher("login.html").include(request, response);
        }
    }
}
```

Java Compiled File (LogoutServlet.class) :

```
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;
import java.io.IOException;
```

```
@WebServlet({"/logout"})
public class LogoutServlet extends HttpServlet {

    public LogoutServlet() {
    }

    protected void doGet(HttpServletRequest var1, HttpServletResponse var2)
            throws ServletException, IOException {

        HttpSession var3 = var1.getSession(false);

        if (var3 != null) {
            var3.invalidate();
        }

        var2.sendRedirect("login.html");
    }
}
```

File (welcome.jsp) :

```
<%@ page session="true" %>
<%
    String user = (String) session.getAttribute("username");
```

```
    if (user == null) {
        response.sendRedirect("login.html");
        return;
    }
%>

<!DOCTYPE html>
<html>
<head>
    <title>Welcome</title>
</head>
<body>
    <h2>Welcome <%= user %>!</h2>
    <p>You are successfully logged in.</p>

    <a href="logout">Logout</a>
</body>
</html>
```

RESULT:
Thus, A simple login system was created where the user's session is tracked by a servlet to demonstrate session tracking using HttpSession.